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HypoSelectSimRAG-Enhancing Answer Accuracy in RAG via Multi-Path Self-Consistent Query Translation

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***Abstract*—Retrieval-Augmented Generation (RAG) enhances the generative capability of large language models by incorporating external knowledge, yet it remains vulnerable to retrieval noise and unstable responses. To address this challenge, we propose HypoSelectSimRAG, a robust framework that integrates multi-path virtual document generation with selective aggregation. Specifically, our method leverages large language models to reformulate the input query into multiple semantically diverse virtual documents, thereby broadening retrieval coverage. The retrieved results are then evaluated through vector similarity with the query, while a self-consistency mechanism is employed to aggregate virtual answers and produce the final output. Experimental results on benchmark datasets demonstrate that HypoSelectSimRAG significantly outperforms standard RAG and multiple baselines in terms of factual correctness and answer faithfulness. The main contributions of this work are as follows: (1) two effective strategies for virtual document generation; (2) several novel methods for multi-path document selection, with the optimal approach identified; (3) empirical verification of the role of self-consistency in virtual document generation; and (4) comprehensive experimental validation showing improved stability and reliability. Overall, HypoSelectSimRAG provides a general framework for building more robust and dependable knowledge-augmented generation systems.**

Keywords: retrieval-augmented generation (RAG),self-consistency, large language model

I. INTRODUCTION

In recent years, the development of large language models (LLMs) has advanced rapidly. Representative models such as GPT [1] and LLaMA [2] have demonstrated strong capabilities in natural language understanding and generation, making human-computer interaction and knowledge acquisition more convenient. However, LLMs still face non-negligible limitations. On the one hand, their knowledge is constrained by training corpora and cutoff dates, making timely updates difficult. On the other hand, due to their massive parameter scales and high training costs, their adaptability in highly specialized domains such as medicine and law remains limited. Moreover, LLMs are prone to generating "hallucinations" [3]—content that appears plausible on the surface but is factually incorrect.

To mitigate these issues, Retrieval-Augmented Generation (RAG) has gradually gained wide attention as an effective paradigm [4]. RAG integrates external knowledge retrieval with generative models, no longer relying solely on parameterized memory. Instead, it allows models to dynamically access databases, knowledge bases, or document collections, thereby significantly improving factual accuracy and domain adaptability. Compared with traditional fine-tuning methods, RAG offers greater flexibility and cost efficiency, since introducing new knowledge only requires updating retrieval indices rather than retraining the model.

Despite its simplicity and efficiency, RAG still faces challenges in practical applications. As the scale of knowledge bases continues to grow, queries are often expressed in only one or two sentences with limited semantic information [5], making it difficult to guarantee the retrieval of sufficient and relevant documents. This in turn, affects the accuracy and completeness of generated answers. Against this backdrop, the HyDE method was proposed. It generates a "hypothetical document", also known as a virtual document. [6] for each query using LLMs to expand the query semantics, and then performs retrieval based on this augmented representation to acquire more relevant documents. However, HyDE also suffers from limitations: its generated virtual documents may still carry hallucinations and redundant information from the LLM [3], leading to noisy retrieval results and ultimately compromising generation quality.

To address these challenges, this paper focuses on the optimization of virtual documents, aiming to retain their semantic expansion advantage while reducing hallucination risk and improving generation quality. To this end, we propose a multi-path RAG optimization framework, HypoSelectSimRAG, which incorporates few-shot prompting, question-oriented prompting [7], and self-consistency [8]. Specifically, we design multiple generation paths for virtual documents to achieve model self-consistency, thereby alleviating hallucination issues. Among these paths, some adopt few-shot prompting, providing the model with a small number of examples to learn generation patterns; others employ question-oriented prompting, classifying the query first and then selecting an appropriate template based on its category to generate more targeted virtual

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documents. Furthermore, for the multiple generated virtual documents, we introduce five selection strategies, ultimately adopting the Best Vector approach—choosing the virtual document with the highest cosine similarity to the original query in the semantic space as the final retrieval basis. By optimizing selection among multiple virtual documents, this approach effectively enhances document relevance and subsequently improves the quality of RAG-generated answers.

Experiments were conducted on the rag-dataset-1200 dataset from the HuggingFace platform, with systematic evaluation based on the RAGAS framework [9]. The core evaluation metric was Factual Correctness (F1). Experimental results demonstrate that both few-shot prompting and question-oriented prompting significantly improve the quality of virtual documents, while the multi-path selection mechanism effectively reduces hallucination rates, thereby enhancing the factual consistency and reliability of RAG's generated answers. Combining these techniques yields substantial improvements in both the accuracy and robustness of RAG.

A. Related Work

This chapter introduces related research from four perspectives: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Query Expansion, and Self-Consistency.

1) Transformer-based Large Language Models

In recent years, Large Language Models (LLMs) based on the Transformer architecture have demonstrated remarkable performance in natural language generation and understanding tasks. The introduction of the Transformer model has revolutionized the paradigm of sequence modeling by overcoming the long-range dependency issues of traditional RNNs through its self-attention mechanism [10]. Autoregressive pre-trained models such as the GPT series excel in language generation tasks, while BERT [11], with its bidirectional encoding structure, significantly enhances semantic understanding, showing strong generalization and contextual comprehension capabilities [1].

As model size increases, GPT-3 has shown impressive few-shot learning capabilities [12]. Subsequent models, such as PaLM series have further demonstrated that the quality of training data is critical for parameter efficiency [13]. Current research focuses on two main directions: (1) optimization of the attention mechanism, such as sparse and linear attention techniques [14]; (2) innovation in training paradigms, such as improving generation quality through Reinforcement Learning with Human Feedback (RLHF) [15]. These advancements continue to push the performance boundaries of LLMs and make them increasingly applicable in practical scenarios.

2) Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) introduces external document retrieval modules to inject "non-parametric knowledge" into the "parametric generation" process, effectively extending the knowledge boundaries of language models [4].

Retrieval modules have developed into a comprehensive technical system: starting from term frequency-based methods such as BM25 [16], evolving to dense retrieval approaches like DPR and Contriever based on pre-trained semantic vectors [5], and eventually to hybrid retrieval strategies such as ColBERT, which combine the advantages of sparse and dense retrieval [17]. Generation-side fusion strategies have also diversified: Fusion-in-Decoder (FiD) integrates documents through cross-attention at the document level [18]; REPLUG incorporates documents during the generation process for post-hoc refinement [19]; and Self-RAG introduces a learnable router module that dynamically decides whether to retrieve based on task complexity [20].

However, there remain critical bottlenecks: if the retrieval module introduces irrelevant documents or noisy redundancy, it can mislead the generation process and degrade answer quality. Therefore, recent research focuses on enhancing overall system robustness through context awareness, trustworthy document filtering, and retriever optimization [6]. Yet, the retrieval pipeline is still vulnerable to noisy evidence and limited query formulation, which directly motivates research on Query Expansion.

3) Query Expansion

In retrieval-augmented generation (RAG), the core idea of Query Expansion is to generate diverse queries or sub-questions in order to cover a broader semantic space and improve the recall of relevant documents. This strategy alleviates the limitations of relying on a single query formulation and has become an important means of enhancing the performance of RAG [21].

Among existing approaches, Reciprocal Rank Fusion (RRF) is often regarded as a representative method of query expansion, which aggregates retrieval results from multiple expanded queries with weighted fusion to produce a more robust candidate set [22]. Building on this idea, RAG-Fusion further leverages large language models to generate multiple complementary queries and integrates their results using mechanisms similar to RRF, thereby improving both the coverage and stability of retrieval [23].

In parallel, more structured expansion strategies have also been explored. For example, the Decomposition approach breaks down complex questions into sub-questions, which are then retrieved and reasoned over separately, showing strong effectiveness in multi-hop question answering [24]. Similarly, Step-back Prompting generates queries by first abstracting the question into higher-level concepts and then refining back to the original question, which enhances the robustness of query expansion [25].

Overall, these methods have achieved notable progress in improving recall and robustness, but certain limitations remain. For instance, expanded queries often face a trade-off between diversity and relevance, and some generated variants may deviate from the user's original intent, thereby weakening the final retrieval performance [26]. Consequently, how to ensure close alignment between expanded queries and the original question while maintaining semantic diversity remains an important direction for further research.

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4) Multi-Path Reasoning and Self-Consistency

To address the instability and unreliability of model outputs in complex reasoning tasks, researchers have proposed Multi-Path Reasoning and Self-Consistency [8]. This approach suggests that for a given question, generating multiple independent reasoning paths and selecting the optimal answer through voting, similarity aggregation, or scoring can effectively reduce output variance and improve accuracy.

Representative methods such as ReAct [27]and Tree-of-Thoughts construct structured reasoning path spaces, enabling the model to possess stronger traceability and exploration capabilities [28]. The Reflexion system introduces a post-generation evaluation phase, allowing the model to self-correct based on intermediate feedback [29]. Strategies like Self-Refine further guide the model through multi-round generation and rewriting to improve the consistency of candidate outputs[30]. In multi-hop QA tasks like HotpotQA, such methods consistently yield a 5 – 9 percentage point increase in final accuracy.

It should be clarified that although the principle of self-consistency[8] can be broadly applied to guide multi-step reasoning , in the proposed HypoSelectSimRAG framework we primarily employ it in the *post-generation selection* and aggregation stage, rather than directly constraining the generation of each individual virtual document. By producing multiple diverse candidate documents and subsequently applying a consistency-driven strategy to identify the most reliable one, our approach mitigates the noise and hallucinations that may arise from a single generation path, thereby improving the accuracy of subsequent retrieval. This application explicitly situates self-consistency [8] in the stage of *post-generation selection* rather than in single-path generation.

II. Model

The retrieval-enhanced generation (RAG) model proposed in this paper (HypoSelectSimRAG) mainly consists of the following 3 components: (1) Generation of multi-path reasoning branches based on the document content; (2) Design of question-oriented prompt templates and few-sample data prompt templates; (3) Merging and summarizing of multiple branch answers and generating the final answer. The main structure of the model is shown in Figure 1.

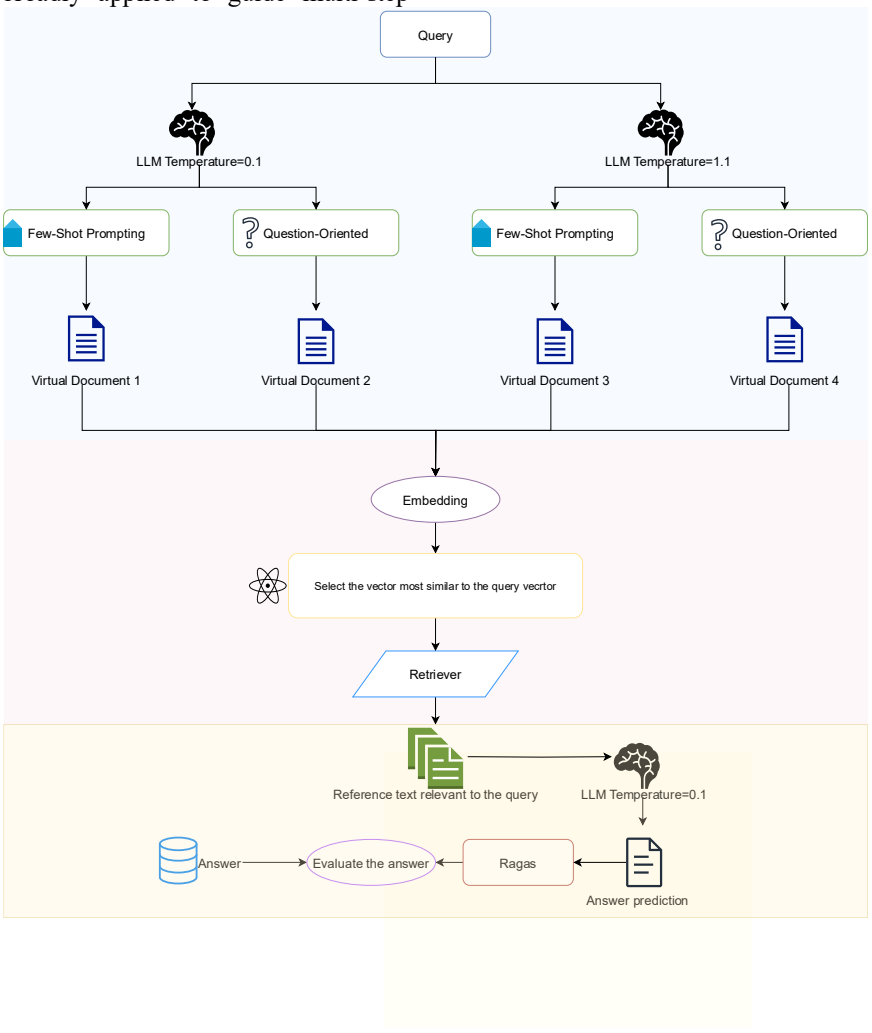


Figure 1 Overall Architecture of the Mode. Phase I: The LLM generates four Virtual Documents using Few-shot and Question-oriented methods under different temperature settings. Phase II: The four documents are vectorized, and the one with the highest semantic similarity to the query is selected for retrieval. Phase III: The LLM generates an answer based on the query and the retrieved document, which is then evaluated against the ground-truth answer using RAGAS.

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The HypoSelectSimRAG model combines the advantage of output diversity in large language models (LLMs). To enhance the diversity of generated responses, diversified prompt strategies are applied. In addition to the basic functions required by conventional RAG models, such as incorporating documents relevant to the query, HypoSelectSimRAG introduces a small number of question–answer examples to improve generalization. After multiple virtual documents are generated by the LLM, the model performs selection and integration of these responses, and the combined result is embedded into vectors for document retrieval. Finally, the LLM produces the final answer based on the retrieved contextual information.

In this section, we first describe the construction of virtual documents, followed by the introduction of the few-shot and question-oriented approaches. We then present five multiple retrieval strategies: Self-Consistency Voting, Self-Consistency Score, Self-Consistency Summary, Average Vector, and Best Vector. Through experimental validation, we compare their performance and ultimately adopt Best Vector as the core method of the HypoSelectSimRAG framework.

A. Virtual Document

Then, In HypoSelectSimRAG, the Virtual Document is employed to generate candidate answers for different branches. The core of the Virtual Document can be summarized as a two-stage zero-shot retrieval process. First, a large language model (LLM), such as InstructGPT, generates a virtual document in response to the query. Although this document may contain certain factual inaccuracies, it is able to emulate the linguistic style and content patterns of relevant documents. Subsequently, an unsupervised contrastive learning encoder encodes the Virtual Document into a dense vector representation, which is then used to perform approximate nearest neighbor search within the document corpus. The dense bottleneck of the encoder effectively filters out part of the hallucinated details while preserving semantic signals that are highly relevant to real documents. This approach fully leverages the generalization capability of LLMs and the document–document similarity space provided by existing encoders, without relying on any relevance labels or additional training. As a result, it enables efficient document retrieval in a completely label-free setting.

B. Strategy of Virtual Document Generation

Compared with the previously proposed single-path zero-shot method, our approach extends the framework by introducing two single-path variants: question-oriented virtual document generation and few-shot virtual document generation. These extensions not only improve the accuracy of virtual document construction but also enrich the diversity of document branches.

1) Virtual Document Generation oriented towards Question Classification

The Virtual Document Generation method oriented towards Question Classification, hereafter referred to as question-oriented, introduces an LLM-based question classifier to optimize both the retrieval and generation processes. In this system, user queries are first fed into the classifier, which categorizes them into predefined types such as extractive QA, complex reasoning, fact-checking, and conversational QA, and outputs the corresponding category codes. This classification step is crucial, as it determines the prompt template employed in the subsequent retrieval and generation stages. Based on the classification results, the system selects a matching template from a predefined library to generate a virtual document. Although fictional, this document effectively captures semantic patterns relevant to the query. The generated document is then embedded into a dense vector space to retrieve the most relevant fragments from the real document corpus. These retrieved fragments, combined with the original query, are subsequently provided to the LLM, enabling it to generate more accurate and detailed answer. The overall workflow is illustrated in Figure 2.

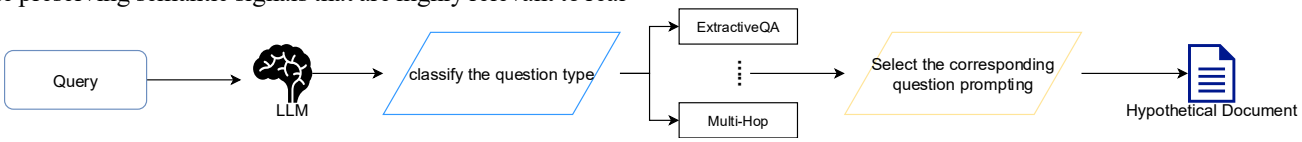


Figure 2 Workflow of the Question-Oriented Method

2) Virtual Document Generation with Few-Shot Data

The implementation of the Virtual Document Generation method with Few-Shot Data is similar to that of the question-oriented virtual document generation approach. This method incorporates a small number of question–answer pairs into the prompts, allowing the large language model (LLM) to learn from them, thereby optimizing the content of the virtual document and improving the accuracy of the answers. Unlike the previously mentioned question-oriented approach, this method does not introduce additional LLMs throughout the process. Instead, it leverages known sample data only during the stage of generating the virtual document, guiding the model to learn similar response patterns from the samples and apply them to the document being generated. This not only enhances the quality of the generated content but also broadens the semantic representation of the query. The generated virtual document is then embedded into a vector space and used to retrieve relevant fragments from the real document corpus. The overall workflow is illustrated in **Figure 3.**

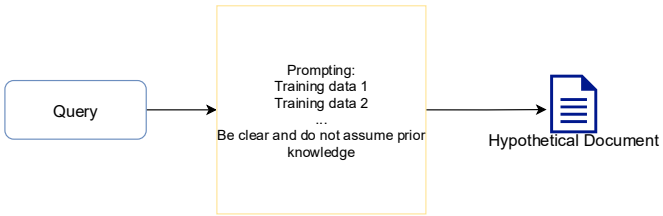


Figure 3 Workflow of the Few-Shot Method

C. Multi-Path Selection for Virtual Documents

The multi-path selection branch design for Virtual Documents originates from the two generation methods proposed earlier. After implementing these two structures, we fine-tuned the corresponding LLM parameters, resulting in four

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configurations: a Question-Oriented LLM with a temperature of 0.1, a Question-Oriented LLM with a temperature of 1.1, a Few-Shot LLM with a temperature of 0.1, and a Few-Shot LLM with a temperature of 1.1. The choice of temperature parameters (0.1 and 1.1) was deliberate, aiming to capture two extremes of the generation spectrum and thereby maximize path diversity.

Low temperature (T = 0.1): A lower setting biases the LLM toward more deterministic and conservative outputs. Virtual documents generated under this condition tend to contain safer, factually grounded statements that closely align with the model's training distribution, thus reducing the risk of hallucinations.

High temperature (T = 1.1): A higher setting encourages the model to generate more creative and exploratory outputs. This helps produce semantically richer documents that may capture implicit aspects of the query, albeit with a higher likelihood of introducing inaccuracies.

By combining these two contrasting generation styles, our framework ensures that the candidate pool includes both "precise" and "diverse" options, providing high-quality and comprehensive inputs for the subsequent self-consistency selection step. While a broader ablation study across additional temperature values could further refine this choice, these two settings effectively represent the balance between precision and diversity that underpins our multi-path strategy.

To determine the final version of the virtual document from the four candidates generated through different paths, we design five combination or preference strategies. These strategies constitute the core mechanism through which our framework realizes self-consistency [8]: by comparing and integrating multiple independently generated results, the model is able to "self-calibrate" and select the document that is most consistent with the intent of the original query and most reliable in content. The five strategies are as follows: Self-Consistency Voting, Self-Consistency Summary, Self-Consistency Scoring, Average Vector, and Best Vector.

Self-Consistency Voting is the earliest method, relying on a voting mechanism based on cosine similarity to select the optimal Virtual Document. After obtaining the four candidate outputs, we compute their pairwise cosine similarity in a matrix form. Each response casts votes according to its similarity with others, and the answer receiving the most votes is selected as the final document.

Self-Consistency Summary aggregates multiple candidate Virtual Documents into a single unified version. Using a prompt template, the LLM is guided to summarize the answers into a more accurate and concise document.

Self-Consistency Scoring is similar to the summary method but instead positions the LLM as a scoring system. It evaluates the four candidate documents against the original question in terms of factual accuracy, relevance, and completeness, and then selects the best one.

Average Vector differs from the previous two approaches. In this method, the four outputs and the original question are jointly embedded into a vector space, and their vectors are averaged. The resulting average vector is then directly used for context retrieval in the database.

Best Vector aims to directly select the most semantically aligned Virtual Document. The four candidates are embedded into a vector space, and their cosine similarity with the original question is computed. The document with the highest similarity score is selected, and its corresponding vector is used for subsequent context retrieval.

III. Experimental Setup

This section presents the experimental setup of this study, including the datasets used, implementation details, baseline models for comparison, and evaluation methods.

A. Dataset

The experiments in this study are conducted on the rag-dataset-1200, which was released on HuggingFace by NeuralBridge under the Apache 2.0 license. This dataset is designed to facilitate research on optimizing Retrieval-Augmented Generation (RAG) models and consists of 1,200 samples, each containing three fields: context, question, and answer. Specifically, the context is extracted from the Falcon-RefinedWeb corpus, the question corresponds to the given context, and the answer provides the ground-truth response. The dataset is partitioned into training and test sets. In our experiments, the test set serves as the primary benchmark for evaluating the proposed RAG pipeline, while a small subset of the training set is selectively utilized for few-shot prompting in certain settings. Due to computational resource limitations, we randomly sample 200 instances from the test set for the final experiments.

B. Implementation Details

In our experiments, OpenAI's GPT-3.5-turbo was adopted as the backbone large language model. For the virtual document generation module within the RAG framework, the temperature parameter was set to either 0.1 or 1.1, depending on the specific branch and task requirements, in order to enhance the diversity of generated virtual documents. In contrast, for the generation module of the overall RAG pipeline, the temperature was fixed at 0.1. To avoid output truncation, no upper limit was imposed on the max_tokens parameter.

For the retrieval process, we empirically determined that setting chunk_size to 512 and overlap to 100 yields superior retrieval performance on the given dataset. The top-k value for nearest neighbor retrieval was set to 5, ensuring that relevant document segments were effectively retrieved.

Regarding embedding strategies, both OpenAI's text-embedding-ada-002 (dense retrieval) and BM25 (sparse retrieval) were employed for comparative analysis, with the goal of identifying the most appropriate retrieval strategy for the proposed pipeline. In the dense retrieval setting, cosine similarity was utilized as the similarity metric.

C. Baseline

To comprehensively evaluate the performance of the proposed HypoSelectSimRAG model, we designed two categories of baselines in our experiments: comparative baselines and ablation baselines. The former is employed to

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assess the effectiveness of our approach against existing mainstream methods, while the latter is used to analyze the contribution of each individual component to the overall model performance.

1) Comparative Baselines

We select the following models as comparative baselines:

(1) Standard RAG model: The most fundamental retrieval-augmented generation approach, which retrieves documents solely based on the query and directly feeds the retrieved content into the large language model for answer generation, without applying any special prompting strategies.

(2) HyDE (Hypothetical Document Embeddings): A retrieval method based on hypothetical document generation, which generates query-related hypothetical documents for embedding, thereby improving semantic retrieval effectiveness[6].

(3) RRF (Reciprocal Rank Fusion): A commonly used multi-retriever fusion method that combines retrieval results from different sources via reciprocal rank weighting to enhance retrieval robustness[22].

(4) Multi-path integration methods: Approaches that introduce multiple reasoning paths during retrieval and generation, with different path selection strategies, including:

- Self-Consistency Voting
- Self-Consistency Score
- Self-Consistency Summary
- Average Vector
- Best Vector

2) Ablation Baselines

To examine the importance of each component in HypoSelectSimRAG, we further designed a series of ablation experiments, including:

(1)FewShot (temperature = 0.1)

(2)FewShot (temperature = 1.1)

(3)Question-Type(question-oriented) (temperature = 0.1)

(4)Question-Type (question-oriented) (temperature = 1.1)

These four experiments are designed to evaluate the performance of a single reasoning path under different prompting strategies and temperature settings.

(5)HypoSelectSimRAG w/o Few-Shot (FS)

(6)HypoSelectSimRAG w/o Question Type (QT)

These two experiments are conducted to analyze the performance differences after removing the Few-Shot prompting and question-typing modules, thereby validating the necessity of these design components.

In summary, the baseline design in this section is complementary: comparative baselines are employed to verify the external competitiveness of the proposed method against representative retrieval and path-selection approaches, while ablation baselines are used to quantify the marginal contribution and necessity of each internal component. Specifically, the single-path settings (Hypo-FewShot and Question-Oriented Prompting under different temperatures) are derived as single-branch variants from the complete four-path architecture, whereas "w/o FS" and "w/o QT" are obtained by removing the corresponding modules to assess their impact. To ensure fairness, all experiments adopt the same data partitioning, consistent retrieval and generation hyperparameters (see Section 3.2), and unified evaluation protocol (see Section 3.4, RAGAS).

D. Evaluation Methods

This study adopts Ragas (Retrieval-Augmented Generation Assessment) as the primary evaluation framework. Ragas addresses the multidimensional evaluation challenges of RAG architectures and introduces the following metrics:

(1)Factual Correctness (F1): Assesses the consistency between the generated result and the ground-truth answer.

(2)Context Recall: Measures the coverage of the retrieved context with respect to the target answer.

(3)Faithfulness: Evaluates whether the generated answer remains faithful to the retrieved context.

In our experiments, Factual Correctness (F1) is taken as the core metric, while Context Recall and Faithfulness serve as auxiliary metrics to facilitate further analysis and optimization.

IV. Experimental Results

In this chapter, we conduct systematic experiments and evaluations on the proposed HypoSelectSimRAG model as well as its various baseline methods.

A. Main Experimental Results

We first evaluated the implementation performance of HypoSelectSimRAG on GPT-3.5-turbo and compared it against the baseline methods on the rag-dataset-1200 benchmark. The evaluation was conducted using the RAGAS framework. The experimental results are presented in Table 1, and based on the observed outcomes, we derive the following conclusions:

TABLE 1 MAIN EXPERIMENTAL RESULTS

Method	**Factual Correctness (F1)***	**Context Recall**	**Faithfulness**
RAG(dense)	0.5257	0.8640	0.8537
RAG-BM25	0.5130	0.9205	0.8949
HyDE(dense)	0.6178	0.8422	0.8160
HyDE-BM25	0.6578	0.9027	0.8385
Self-Consistency Voting-BM25	0.6553	**0.9404**	0.8619
Self-Consistency Voting (dense)	0.6786	0.9225	0.8753
Self-Consistency Score-BM25	0.6757	0.9159	0.8743
Self-Consistency Score (dense)	0.6623	0.9343	0.8879
Self-Consistency Summary-BM25	0.6647	0.9272	0.8709

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Self-Consistency Summary (dense)	0.6886	0.9361	0.8588
Self-Consistency RRF-BM25	0.6287	0.8568	0.8241
Self-Consistency RRF (dense)	0.6443	0.8785	0.8428
Average Vector (dense)	0.6738	0.9354	0.8830
HypoSelectSimRAG(Best Vector)-BM25	0.6740	0.9305	0.8722
HypoSelectSimRAG(Best Vector) (dense)	**0.6910**	0.9395	**0.8965**

Note: Factual Correctness (F1) is the primary evaluation metric in this study, as it directly reflects the reliability of the generated content.

(1) Incorporating virtual document–based retrieval methods leads to a significant improvement in accuracy compared with the traditional RAG pipeline. This effectively alleviates the inherent issues of original RAG queries, which are often short, vague, and lacking in information. However, the introduction of virtual documents also increases the volume of retrieved content, thereby introducing certain irrelevant or redundant information. As a result, some retrieval-related metrics (e.g., Context Recall) show a slight decline.

(2) The overall model consistently maintains a high level of performance on the Faithfulness metric. This metric evaluates whether the generated answers are strictly grounded in the retrieved context without introducing irrelevant or contradictory information. A high level of Faithfulness ensures the reliability of LLM outputs to a considerable extent, thereby strengthening the credibility of the experimental findings.

(3) Multi-path selection methods demonstrate remarkable effectiveness in mitigating the redundancy issue introduced by virtual documents in HyDE[6]. Leveraging the principle of self-consistency, multi-path selection reduces hallucinations and redundancy. Compared with the standard HyDE[6], the five methods—Self-Consistency Voting, Self-Consistency Score, Self-Consistency Summary, Average Vector, and HypoSelectSimRAG (Best Vector)—all achieve better results in Context Recall, indicating that redundancy is effectively suppressed. Moreover, they significantly improve accuracy in dense retrieval and also show performance gains in most metrics under sparse retrieval, with only a few remaining unchanged.

(4) In dense retrieval settings, the original HyDE method shows relatively low accuracy, whereas multi-path virtual document generation methods substantially enhance performance. This is because dense retrieval heavily depends on semantic vector representations: if HyDE-generated virtual documents contain hallucinations or semantic shifts, retrieval is severely misled. In contrast, multi-path selection mitigates hallucinations and semantic drift, thereby improving accuracy. On the other hand, sparse retrieval is primarily based on keyword matching, which is less sensitive to semantic drift; thus, the original HyDE [6]already performs relatively well, and further optimization yields only limited gains.

(5) In the accuracy comparison for dense retrieval, among the five methods (Self-Consistency Voting, Self-Consistency Score, Self-Consistency Summary, Average Vector, and HypoSelectSimRAG [Best Vector]), Best Vector and Self-Consistency Summary perform most prominently. We attribute the advantage of Best Vector to its ability to select virtual documents most closely aligned with the query, thereby significantly enhancing the specificity and relevance of retrieved content while reducing redundancy. Meanwhile, Self-Consistency Summary integrates the core information from four reasoning paths, effectively avoiding redundancy while preserving high-quality semantic features. Although Average Vector is simple and robust, it may dilute strong signals. Self-consistency voting can mitigate extreme outputs to some extent, but it does not fully account for direct query relevance, making it less effective than Best Vector. Self-Consistency Score relies heavily on the LLM's sensitivity to prompts and surface-level linguistic cues, lacking a deeper semantic alignment with the query, which limits its effectiveness.

To illustrate more concretely why the Best Vector strategy achieves superior performance, consider the query: *"What were the economic impacts of the 2020 Olympic Games being postponed?"*

In our four-path setting, a high-temperature path might produce a creative but slightly off-topic virtual document focusing more on cultural impacts, whereas a low-temperature path might generate a conservative document restricted to ticket revenue losses.

The Self-Consistency Summary strategy would attempt to merge these documents, potentially yielding a more comprehensive but diluted representation in which the central theme is less salient.

The Average Vector strategy would average their vector embeddings, which could shift the final representation away from the core theme of economic impacts, resulting in a more balanced but less precise signal.

Self-consistency voting might incorrectly prioritize a document that happens to share superficial similarities (e.g., multiple documents mentioning "Tokyo") rather than the one most relevant to the economic dimension.

In contrast, the Best Vector strategy directly measures the semantic alignment between each complete virtual document and the original query. It is therefore most likely to select the document that sharply focuses on ticket revenue losses, as its embedding achieves the highest cosine similarity with the query. This direct alignment with the user's intent ensures both high relevance and precision in retrieval, which explains its superior performance on the Factual Correctness metric.

In sparse retrieval, the Self-Consistency Score performs relatively better, likely because sparse retrieval is more tolerant of deviations in LLM-generated virtual documents. Nonetheless, overall, HypoSelectSimRAG (Best Vector) demonstrates the best performance across all three evaluation metrics—Factual Correctness, Context Recall, and Faithfulness—particularly on the most critical metric, Factual Correctness. Therefore, we ultimately select it as the optimal RAG pipeline for this study.

B. Ablation Study

1) Ablation Study 1: Single-Path

To verify the effectiveness of the multi-path selection strategy and its underlying principle of model self-consistency in virtual document generation, we conducted experiments on

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four single-path models within the longitudinal ablation baselines, namely Few-Shot (temperature = 0.1 / 1.1) and Question-Oriented (temperature = 0.1 / 1.1). On the rag-dataset-1200 dataset, we evaluated the performance using Ragas metrics under dense retrieval and compared the results against HypoSelectSimRAG. The experimental results are presented in Table 2.

TABLE 2 ABLATION STUDY 1 RESULTS

Method	Factual Correctness (F1)*	Context Recall	Faithfulness
Few-shot (T=0.1)	0.6646	0.9129	0.8674
Few-shot (T=1.1)	0.6560	0.9029	0.8670
Question-type (T=0.1)	0.6640	0.9201	0.8622
Question-type (T=1.1)	0.6647	0.9205	0.8631
HypoSelectSimRAG	**0.6910**	**0.9395**	**0.8965**

The results show that the single-path ablation model performs significantly worse than the full model across all metrics, with the poorest performance observed in Factual Correctness (F1). This indicates that virtual documents generated by a single path are of limited quality and prone to introducing hallucinations or redundant information. In contrast, the multi-path generation and selection strategy adopted by HypoSelectSimRAG not only integrates diverse information sources but also effectively selects virtual documents most aligned with the query through a filtering mechanism, while reducing redundancy and hallucinations by leveraging the principle of self-consistency. Moreover, this experiment indirectly demonstrates the robustness of the multi-path strategy: even when certain paths produce low-quality outputs, the model can still identify high-quality content through the self-consistency mechanism.

2) Ablation Study 2: Multi-Path Single-Strategy

We further evaluated the roles of Few-shot Prompting (FS) and Question-type Prompting (QT) through two ablation settings: (1) w/o FS, keeping only QT; and (2) w/o QT, keeping only FS. Both were tested under the same configurations. Results (Table 3) show that removing either strategy degrades performance, especially on Factual Correctness (F1). This indicates that FS and QT are complementary: FS improves task understanding via few-shot examples, while QT provides stronger task-specific guidance. Their combination enhances path diversity and supports self-consistency, leading to more accurate and comprehensive results.

TABLE 3 ABLATION STUDY 2 RESULTS

Method	Factual Correctness (F1)*	Context Recall	Faithfulness
HypoSelectSimRAG w/o FS	0.6611	0.9222	0.8707
HypoSelectSimRAG w/o QT	0.6605	0.9315	0.8361
HypoSelectSimRAG	**0.6910**	**0.9395**	**0.8965**

In summary, the experimental results demonstrate that different retrieval methods and path selection strategies exhibit significant performance differences. Under sparse retrieval conditions, the Self-Consistency Score shows certain advantages, whereas the HypoSelectSimRAG (Best Vector) combined with dense retrieval achieves overall superior performance. In particular, it outperforms all other approaches on the most critical metric—Factual Correctness—while also delivering strong results in Faithfulness and Context Recall. Therefore, this study ultimately identifies HypoSelectSimRAG (Best Vector) as the optimal RAG pipeline for subsequent analysis and applications.

V. CONCLUSION

In this work, we presented HypoSelectSimRAG, a novel framework that enhances Retrieval-Augmented Generation (RAG) by combining multi-path query reformulation with self-consistent answer selection. Unlike conventional RAG models that rely on a single query and are thus sensitive to retrieval noise, our method generates semantically diverse query variants to broaden retrieval coverage and mitigate information sparsity. A self-consistency mechanism is further employed to aggregate multiple candidate answers and select the most reliable response based on semantic similarity.

Experimental results demonstrate that HypoSelectSimRAG consistently outperforms baseline RAG systems in terms of factual correctness and context recall. These findings confirm the effectiveness of multi-path and self-consistency strategies in enhancing the stability and accuracy of knowledge-intensive question answering. In future work, we plan to investigate adaptive path selection and reinforcement learning-based optimization to further improve retrieval robustness. Overall, HypoSelectSimRAG provides a promising direction for building reliable and scalable RAG systems in real-world applications.

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APPENDIX A: QUESTION-ORIENTED PROMPT TEMPLATES

To improve the quality of virtual documents, HypoSelectSimRAG employs question-oriented prompting in two stages:

1. Query Classification Prompt

The following prompt is used to assign each query to one of the predefined categories (A–N). If the type is unclear, the model outputs "X."

```
Prompt:
You will receive a user question. Please select the most appropriate type from the following tasks and directly output the category letter:

A. ExtractiveQA (Draw answers from the given document)

B. Complex/LongQA (Multi-jump reasoning or long responses are required)

C. FactCheck (Judge authenticity)

D. Conversation (Casual chat or instructions)

E. Multi-hop Reasoning (Cross-document logical links required)

F. TemporalQA (Time-sensitive answers)

G. MultimodalQA (Text + charts / images / code / video)

H. ComparativeQA (Compare entities or options)

I. CounterfactualQA (Hypothetical or "what-if" scenarios)

J. NumericalQA (Calculations, statistics, trend analysis)

K. Long-doc Summarization (Condense very long documents)

L. Domain-term Explanation (Define specialized jargon in context)

M. Implicit-intent QA (Surface question hides deeper need)

N. Real-time QA (Answers depend on live data feeds)

If you are NOT SURE, please output 'X'.
```

2. Category-Specific Templates

After classification, the query is inserted into the corresponding template.

Prompt:

```
"A. ExtractiveQA": (
    "We are looking for a concise excerpt from a document that directly answers the question:\n"
    "Question: {question}\nPassage:"
),
"B. Complex/LongQA": (
    "Provide a comprehensive, multi-paragraph answer that covers all aspects of:\n"
    "Question: {question}\nAnswer:"
),
"C. FactCheck": (
    "Generate a short factual statement that either confirms or refutes the following claim:\n"
    "Claim: {question}\nEvidence:"
),
"D. Conversation": (
    "Write a helpful, friendly response in conversational style to:\n"
    "User: {question}\nBot:"
),
"E. Multi-hop Reasoning": (
    "Compose a short paragraph that connects information from multiple sources to answer:\n"
    "Question: {question}\nIntegrated answer:"
```


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```
    ),
    "F. TemporalQA": (
        "Focus on the current or most up-to-date information to answer:\n"
        "Question: {question}\nCurrent answer:"
    ),
    "G. MultimodalQA": (
        "Describe the relevant textual explanation or caption that would accompany the necessary chart/image to answer:\n"
        "Question: {question}\nDescription:"
    ),
    "H. ComparativeQA": (
        "Produce a side-by-side comparison paragraph that highlights similarities and differences for:\n"
        "Question: {question}\nComparison:"
    ),
    "I. CounterfactualQA": (
        "Imagine a hypothetical scenario and explain what would happen if the premise were true:\n"
        "Question: {question}\nHypothetical explanation:"
    ),
    "J. NumericalQA": (
        "Provide a short paragraph that includes any required calculations or numeric reasoning to answer:\n"
        "Question: {question}\nNumeric answer:"
    ),
    "K. Long-doc Summarization": (
        "Write a concise summary that captures the key points needed to answer:\n"
        "Question: {question}\nSummary:"
    ),
    "L. Domain-term Explanation": (
        "Define and briefly explain the specialized term or concept in context:\n"
        "Question: {question}\nExplanation:"
    ),
    "M. Implicit-intent QA": (
        "Infer the underlying need behind the user's surface question and provide a targeted answer:\n"
        "Surface question: {question}\nUnderlying answer:"
    ),
    "N. Real-time QA": (
        "State the most recent data-driven answer that depends on live or real-time feeds:\n"
        "Question: {question}\nReal-time answer:"
    ),
    "X. Model cannot judge": (
        "Provide a general-purpose passage that may help answer:\n"
        "Question: {question}\nPassage:"
```


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